

**ABTS (2,2'-azino-bis-3-ethylbenzthiazoline-6-sulphonic acid) Assay**  
**For use with sea grapes (*Caulerpa lentillifera*)**

Common radical scavenging assay, which is based on the generation of the blue/ green ABTS radical cation (ABTS<sup>•+</sup>), that can be reduced by antioxidants.

The spectrophotometric method shows the radical scavenging ability of a sample.

In order to test the ability, we need to use a standard, here we use Trolox equivalent antioxidant capacity (TEAC, 6-Hydroxy-2,5,7,8-tetramethylchromane-2-carboxylic acid).

**Sample preparation**

- Store and process the samples in darkness. Fresh samples should be frozen (-80°C) for at least 24h prior freeze drying.
- For kelps 0.05 g of dry weight is a good range for measurements in FastPrep tubes; put ceramic ball in tubes
- Pulverize samples in FastPrep with one ball (or by hand), handle the Fastprep with care

**DAY 1:**

**Preparation of stock solutions**

| Short name           | Chemical name                                          | Molar mass            |
|----------------------|--------------------------------------------------------|-----------------------|
| ABTS                 | 2,2'-azino-bis-3-ethylbenzthiazoline-6-sulphonic acid  | 548.7 g/mol           |
| Trolox               | 6-Hydroxy-2,5,7,8-tetramethylchroman-2-carboxylic acid | 50.29 g/mol           |
| Potassium persulfate |                                                        | 270.322 g/mol         |
| Ethanol, absolute    |                                                        | 789 kg/m <sup>3</sup> |

**ABTS-stock solution (7 mM) in 50 mL Falcon tube**

with **25 mL Bidest-H<sub>2</sub>O** and **0.0901 g ABTS (powder – stored in fridge, one bottle has 1 g powder)**.

Pipette 4x5 mL Bidest-H<sub>2</sub>O to wash the ABTS in a Falcon tube

The mixture will be shaken by hand under the fume cupboard (bottle should be dark or wrapped in aluminum foil to protect from light)

then add **0.0166 g of K<sub>2</sub>S<sub>2</sub>O<sub>8</sub> (Potassium persulfate)** in order to have a 2.45 mM solution.

Pipette 1x5 mL Bidest-H<sub>2</sub>O to wash the Potassium persulfate in the Falcon tube

The mixture will be left for at least 16 h until use. The radical is stable for 2-3 days while kept in darkness at room temperature.

**1 L 70% Ethanol**

take **785 mL of 90% v/v ethanol**, or **730 mL of 95% v/v ethanol**, or **707 mL of 99% v/v ethanol**; add distilled or filtered water to make up a volume of 1L; leave to cool and top up with water again to bring the volume back to 1L (mixing water and ethanol together produces a reaction whereby volume is reduced).

**Trolox stock solution in 50 mL Schott bottle**

We need a 2.5 mM Trolox solution in order to make a standard curve.

**25 mL Ethanol (70%)** and **0.0156 g Trolox powder**.

Pipette 5x5 mL Ethanol to wash Trolox in Schott bottle

The solution will be mixed using a magnet stirrer and a magnetic plate overnight.

## DAY2:

We would suggest to plan an extra day for the extraction and the preparation of the working solution. It took 7-8 h to get the working solution stable. If time is pressing, the Extraction can be done on DAY1 and the preparation of the ABTS-working solution on DAY2 before the measurements. However, if done so, plan a lot of time for the preparation of the working solution!!

### Extraction (start early in the morning, around 8 am):

- For the extraction:  
**Kelp / Sea grapes:** add **1 mL Ethanol (70 %)** to **0.05 g DW** of sea grape material and extract in a water bath for **4h at 47°C**. The homogenate needs to be **mixed every hour** using a vortexer.  
**Fucus:** add **2 mL Ethanol (70 %)** to **0.01 g DW** of sea grape material and extract in a water bath for **4h at 47°C**. The homogenate needs to be **mixed every hour** using a vortexer.
- Samples are centrifuged at **2,500 g** and **20°C (room temperature)** for **10 min**.
- Pipette supernatant (e.g. 700 µL) in extra Eppi (to stop reaction); put supernatant in fridge over night

### ABTS - Working solution

During the extraction, the ABTS stock solution can be already prepared  
(I would suggest to plan **at least 2 hours for this**, so that it can be made sure that the solution is really stable and it can be mixed properly).

The stock solution will be diluted with **ethanol (absolute)** to an absorption of  **$0.7 \pm 0.02$  at 734 nm**.

**Make a working solution from the stock solution – Do NOT dilute the stock solution!**  
**The stock solution can be transferred in a small beaker or something.**

The amount of **Ethanol (absolute)** should be chosen based on the amount of samples to be measured. **One sample needs 1 mL of the working solution. Also the amount of samples for the calibration curve needs to be considered.**

Therefore, we should use for example for an amount of **25 samples at least 50 mL Ethanol** in order to have a sufficient supply. The amount of stock solution needed in the Ethanol is always different.

Start adding ABTS to the ethanol 1:100 (e.g. 1.5 mL ABTS stock) on 150 ml EtOH). Wait until the absorption is stable, then add ABTS stock solution until an absorption of 0.7 at 734 nm is reached.

**This should be done slowly and careful and with appropriate amount of waiting and shaking!! (IMPORTANT!!)** We would suggest to wait for an hour if it really is stable.

### DAY3: Spectrophotometric measurements

Check the absorption of the working solution. Probably, it will have changed overnight. Add some ABTS until absorption of 0.7 at 734 nm is reached. Wait for an hour if the absorption is stable.

#### Calibration curve of Trolox equivalents (TE mmol 100g<sup>-1</sup> DW):

We prepare a calibration curve from 0–2.5 mM based on the Trolox stock solution and Ethanol 70 % (as used for extraction).

|                           |      |     |     |     |     |      |
|---------------------------|------|-----|-----|-----|-----|------|
|                           |      |     |     |     |     |      |
| <b>Concentration (mM)</b> | 0    | 0.5 | 1   | 1.5 | 2   | 2.5  |
| <b>Ethanol (µL)</b>       | 1000 | 800 | 600 | 400 | 200 | 0    |
| <b>Trolox (2.5 mM)</b>    | 0    | 200 | 400 | 600 | 800 | 1000 |

When the extraction is done and the **ABTS working solution is stable**, the measurement can start. The spectrophotometer needs to be measured with a “**blank**” which is **Ethanol (absolute)**. It is important to put the cuvette in correctly (**The small opening is where the light goes through**; Arrow where the light is travelling through).

Then we need to press “**Fixed**” and chose the function “**ABTS Niko (734 nm)**”. Then we put a **blank Ethanol, absolute and press blank**. Treat the calibration curve as a sample.

#### Prepare sample for measurement:

For the measurements we add **10 µL sample in the cuvettes (first)** and then **1 mL of the ABTS working solution (second, starts the reaction)** and measure at **734 nm** after an **incubation time of 6 min** on a **UV/ VIS-Spectrophotometer (Thermo Scientific Genesys 140/150, Fisher Scientific GmbH, Germany)**.

We would suggest to work in a pair: One person is preparing the cuvettes, the other is measuring the absorption.

##### Person 1:

1. Minute 0: Pipett 10 µL sample, add 1 mL ABTS working solution
2. Start (infinite) time
3. Put cuvette in darkness (suggestion: empty cuvette boxes)
4. Every full minute: Prepare next cuvette; out in darkness next to the first sample

##### Person 2:

5. Minute 6: measures absorption of sample 1; write down in Labbook
6. Empty cuvette and throw away
7. Every full minute: measure absorption of the next sample.

Like this **1 sample takes 1 minute**. Hence, many samples can be measured in one go. Measuring alone is possible, however, then we would suggest to leave of two minutes between the preparing the samples (keep a veeeery good eye on the watch)

**Waste of liquids and hard stuff.** ABTS and Trolox is toxic. Therefore, we put the plastic stuff contaminated with it in a plastic bag and then in the rubbish and the liquid is going to come in the organische Lösungsmittel Abfall – halogen frei.

**Calculations:**

- Trolox: a standard curve using Trolox (0–2.5 mM) was prepared for every measuring day. Based on the standard curve, the concentration in each sample was calculated in Trolox equivalents in mmol/100g:

$$\text{Trolox Equivalent TE (mmol / 100g)} = -[(A734 - 0.686) / 0.2574] * V_f / 10$$

With A734 being the absorption, the two coefficients being obtained from the equation of the standard curve (e.g.  $y = -0.2574 * x + 0.686$ ;  $R^2 = 0.9867$ )

$V_f$  is the dilution factor, which is calculated as following:

$$V_f = ((0.8 \text{ mL} + 0.05 \text{ g}) / 0.05 \text{ g}) * (1010 \text{ uL} / 10 \text{ uL})$$

Literatur: Floegel et al. 2011; Re et al. 1999; Waldmann 2008; Zakaria 2012